

GAYLORD, MI, USA

WMO: 725407

Lat: 45.013N

Lon: 84.701W

Elev: 407

StdP: 96.53

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04854

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.1	-24.1	0.4	-20.3	-21.4	0.6	-17.3	11.0	-6.0	10.1	-5.7	1.1	60	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.0	20.7	28.4	20.0	27.0	19.2	22.5	27.4	21.5	26.1	20.6	24.8	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	16.6	25.0	20.0	15.5	23.7	19.0	14.5	22.8	68.3	27.2	64.3	25.8	61.0	24.9	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.5	7.6	-27.5	33.0	3.7	1.3	-30.2	34.0	-32.3	34.7	-34.4	35.5	-37.0	36.4		
(5)			-27.7	24.0	3.6	1.0	-30.3	24.8	-32.4	25.4	-34.4	26.0	-37.1	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-7.6	-7.3	-2.2	5.1	12.1	17.1	19.7	18.6	14.9	8.0	1.6	-4.3	(6)
(7)		DBStd	11.00	5.96	5.91	6.62	5.57	5.09	4.08	3.44	3.32	4.55	5.02	5.25	5.28	(7)
(8)		HDD10.0	2426	545	485	382	167	34	2	0	0	9	103	254	444	(8)
(9)		HDD18.3	4543	803	718	637	399	203	70	26	37	122	324	501	703	(9)
(10)		CDD10.0	1103	0	0	4	19	101	215	299	266	155	40	4	0	(10)
(11)		CDD18.3	179	0	0	1	1	11	34	67	46	18	2	0	0	(11)
(12)		CDH23.3	1562	0	0	3	12	117	316	596	360	142	15	0	0	(12)
(13)		CDH26.7	374	0	0	0	1	23	78	154	87	29	2	0	0	(13)
(14)	Wind	WSAvg	3.3	3.6	3.5	3.2	3.5	3.3	3.0	2.9	2.7	3.0	3.4	3.6	3.7	(14)
(15)	Precipitation	PrecAvg	809	56	39	40	62	78	82	73	88	86	82	65	56	(15)
(16)		PrecMax	981	108	67	126	105	124	149	149	178	159	183	122	110	(16)
(17)		PrecMin	665	14	10	2	18	24	13	31	32	23	23	24	11	(17)
(18)		PrecStd	75	25	15	27	22	24	35	31	37	31	37	28	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	9.9	21.0	25.0	29.0	31.2	32.3	31.4	30.0	25.4	18.3	11.3	(19)
(20)			MCWB	5.4	7.1	14.1	14.8	19.0	21.5	22.2	21.3	20.1	17.9	12.0	10.0	(20)
(21)		2%	DB	4.6	6.0	13.3	21.1	26.7	28.8	30.1	29.0	27.1	22.0	14.1	6.2	(21)
(22)			MCWB	3.0	3.5	8.7	12.7	17.4	19.9	21.0	20.4	19.1	16.5	10.3	4.8	(22)
(23)		5%	DB	2.3	2.9	10.0	17.7	24.0	27.1	28.2	27.1	24.6	18.8	11.9	4.0	(23)
(24)			MCWB	1.1	1.3	6.5	10.5	15.5	19.1	20.3	19.6	18.0	14.7	9.0	2.8	(24)
(25)		10%	DB	0.9	0.8	6.9	14.5	21.6	24.9	26.8	25.2	22.7	16.1	9.1	2.4	(25)
(26)			MCWB	0.1	-0.5	3.9	8.4	14.7	17.5	19.4	18.2	17.3	12.6	6.7	1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	7.3	14.6	16.7	20.9	23.2	24.1	23.5	22.2	20.1	13.6	10.8	(27)
(28)			MCDWB	7.0	8.6	19.9	21.3	26.7	28.9	29.7	28.1	26.5	23.7	16.6	11.2	(28)
(29)		2%	WB	3.1	3.7	10.3	14.0	19.1	21.4	22.9	21.8	20.7	17.5	11.4	5.5	(29)
(30)			MCDWB	4.6	5.5	12.6	19.7	24.0	26.9	27.7	26.4	24.4	20.3	13.8	6.0	(30)
(31)		5%	WB	1.2	1.4	6.7	11.7	17.6	20.2	21.8	20.8	19.6	15.0	9.0	3.1	(31)
(32)			MCDWB	2.2	2.9	9.0	16.5	22.1	24.7	26.0	25.1	22.7	18.6	11.4	3.9	(32)
(33)		10%	WB	-0.1	-0.5	4.0	9.0	15.7	19.1	20.7	19.8	18.3	12.8	6.9	1.3	(33)
(34)			MCDWB	0.7	0.6	6.8	13.8	20.1	23.2	24.9	23.4	21.7	15.6	8.9	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	8.5	9.9	10.7	11.7	11.3	11.4	10.9	10.9	8.9	7.1	5.9	(35)
(36)			MCDBR	7.4	9.9	12.4	16.0	14.8	13.4	13.0	13.1	13.0	13.0	11.2	7.5	(36)
(37)			MCWBR	6.3	8.1	8.4	9.6	7.5	6.5	6.0	6.1	6.5	7.8	8.0	6.7	(37)
(38)		5% WB	MCDBR	7.0	8.9	11.3	14.1	12.1	11.0	10.6	10.5	10.4	11.4	10.5	7.0	(38)
(39)			MCWBR	6.3	7.7	8.5	9.6	7.1	6.1	5.8	5.7	6.4	8.0	8.3	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.281	0.286	0.306	0.339	0.365	0.383	0.381	0.372	0.359	0.326	0.294	0.268	(40)	
(41)		taud	2.308	2.286	2.406	2.378	2.363	2.331	2.355	2.375	2.411	2.503	2.524	2.426	(41)	
(42)		Ebn at Noon	830	899	931	922	903	884	882	877	861	849	825	820	(42)	
(43)		Edh at Noon	79	98	101	113	119	124	119	113	100	80	65	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.22	2.14	3.56	4.58	5.54	6.05	6.11	5.19	4.03	2.38	1.31	0.92	(44)	
(45)		RadStd	0.12	0.23	0.39	0.45	0.49	0.42	0.26	0.31	0.36	0.28	0.18	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page